



SIMATS Online Programming Lab

C PROGRAMMING



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CO4 - AT2 - CASE STUDY

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QUESTIONS

Q1

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25 marks

**Q2**

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**Q3**

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**Q4**

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25 marks



4/4 answered

Q1 CO4 - AT2 - CASE STUDY

25 marks

A large enterprise resource planning application is built from multiple source files, including modules for sales, inventory, and finance, and all of these modules must reference a common variable that stores the current fiscal year. The variable is defined once in one source file but must be read and, in some cases, modified by functions compiled separately in other source files. Without proper declaration, each module would either fail to compile or maintain its own inconsistent copy of the fiscal year. The design must ensure a single, consistent value is shared across the entire application.

Question: Identify which storage class allows the fiscal-year variable to be accessed across multiple source files, and explain the role of extern in achieving this shared access.

Your Code

```
1 #include <stdio.h>
2 int fiscalYear = 2025;
3 extern int fiscalYear;
4 void displayFiscalYear() {
5     printf("Sales Module: Current Fiscal :
6 }
7 extern int fiscalYear;
8 void updateFiscalYear(int newYear) {
```

```
9     fiscalYear = newYear;
10    printf("Inventory Module: Fiscal Year
11 }
12 int main() {
13     printf("Finance Module: Fiscal Year in
14     displayFiscalYear();
15     updateFiscalYear(2026);
16
17     displayFiscalYear();
18
19     return 0;
20 }
```



Input

```
extern int fiscalYear;
```

Output



Run



Save